

A PROJECT REPORT ON

TRAINING AND PLACEMENT PORTAL

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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(COMPUTER ENGINEERING)

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ABSTRACT

SCTR's Pune Institute of Computer Technology is well known for Placement records all across the country. Various MNCs participate in PICT's Placement process and hire one of the finest talents across the globe. This wonderful process is well designed and managed by TNP Cell under the distinguished leadership of TPO and excellent volunteering of TNP heads. This process can be enhanced by making a few advancements in it. Our project 'Training and Placement Portal' emphasizes on development of a centralized automated platform for TNP Department, so as to streamline the hiring process. The project tries to minimize various issues faced by students and TNP Department throughout the hiring process. The system allows the aspirants to view job postings along with job details and required criteria to get shortlisted for further hiring process. Our project encompasses a well designed dashboard for TNP Department , so as to keep record of student data, job listing and other required parameters. Additionally , a chat system between TNP and students is developed , incorporating the text summarization feature in it. This ensures a proper communication and resolution of problems faced by students during hiring process. The system also generates personalized notifications and deadline reminders for the aspirants in accordance with online assessments and interviews. Taking all concerns into considerations, we designed an optimal automated hiring portal with benefits both the parties involved in the process.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Training and Placement Department lies at the core of any institute. SCTR's Pune Institute of Computer Technology has embarked great placements for all of its students over a period of time. Our project aims to streamline the hiring process for both students and TNP Department by development of a centralized automated Training and Placement Portal. We emphasize on a single platform for students to view job listings, upcoming companies, shortlisting criteria and various other such parameters. Additionally, we plan to develop a well designed dashboard for administration and keep track of students placement records. Beyond this, we also plan to extend this work in student training activities and continuous technical assessments.

1.2 Motivation

Hiring process is a crucial phase for every student during their final year. During this process, a student not only undergoes mental trauma but also becomes susceptible to emotional trauma that then affects their mental health. In amidst of peer pressure and anxiety, students wish to have a smooth and comforting experience during this tedious process. On the other hand, for TNP Department, it's a tiresome task for maintaining records of all the students and have a continuous communication with each of them. Also, it is difficult to keep in co-ordination with students and companies at the same time. To ease this process, both for students and TNP Department, we plan to implement a centralized automated Training and Placement Portal that benefits both students and TNP Department throughout this tedious hiring process.

1.3 Problem Definition and Objectives

While understanding and collecting details from both students and TNP volunteers, we figured out some of the problems faced by both students and volunteers during the process of hiring. The problem in the current process starts from registration to the time when candidate is finally hired. The process of registration for placement starts with collection of all the documents and details of interested students. These documents and details are maintained in a master sheet by TNP Cell. The authenticity and verification of the documents submitted by students is done by volunteers from all the classes and TNP heads. This is a very tiresome process in which human errors may

occur. For each company, students are expected to fill a separate google form for each and every company. A small error while filling the google form leads to disqualification from companies' placement drive. The google form deadline is within a stipulated time frame, A student failing to fill the google form cannot appear for the company's placement drive. The major problem being no repetitive notifications regarding form deadlines. Students who have active backlogs are not eligible for the company's placement drive. If a student clears the backlog in the process of re-evaluation, the student has to wait a span of 10-15 days till the date TNP Cell updates the master sheet. This results in loss of students' career opportunities.

Company specifies its shortlisting criteria to get a list of eligible students, this criteria is specified by TNP Cell in a textual format in google form description. In Spite of this, ineligible students fill the form resulting in absurd data problems and TNP then again needs to refilter the data and send it to the company. During a company's placement drive, students face a lot of issues related to test platforms, login issues, etc. Students then contact TNP heads regarding the issues, it becomes difficult to handle a heap of queries from students on admin side.

The major objective of this project is to address all these problems and make the process streamline for both the parties involved.

1.4 Project Scope & Limitations

The proposed project lies in the space of hiring domain. The process of hiring starts with registration of candidates who aspire to get place at various companies. It then moves on to verifying the candidates' information and thoroughly reviewing their details. Once scrutiny of application is done, the candidate then will be able to access and apply all the opportunities available. Administrators list job opportunities and publish available notices during this process. The main objective of this project is to streamline the above process and help students and administrators throughout the process. The key features of the project include three way authentication mechanism for candidate registration, proper display of job opportunities alongwith required details, a robust shortlisting based upon shortlisting criteria, personalized notifications and reminder system, a chat interface for smooth communication and a descriptive visual dashboard for accessing and managing student data.

This project is developed by taking into consideration the requirements and issues faced during hiring process at SCTR's Pune Institute of Computer Technology. The overall structure of the project is shaped as per the requirements from higher authorities and other stakeholders. This introduces a rigidity and scalability limitation in the project. The portal cannot be versatile and suitable for hiring process of each and every organization.

1.5 Methodology of Problem solving

The proposed portal is a comprehensive centralized automated system planned for enhancement of hiring process and betterment of both students and administrators. Development of a mobile application for students and web based dashboard for Training and Placement Department is a right approach towards solving the problem. The mobile application developed will be compatible for both Android and IOS users. The dashboard can be viewed from any available browser for managing and accessing student data.

A candidate who aspires to appear for placement process can register on the portal by filling in the appropriate academic , technical and background details. Once filled, the application will be sent to the administrator for scrutiny. After verifying, credentials will be allocated to the candidate. If there exists any discrepancy in the information, admin can notify candidate to correct it. On getting credentials, candidate can view all the job opportunities listed on the portal. Based upon the data filled while registering, candidate can only apply for opportunities based upon whether he/she meets the shortlisting criteria specified. All details regarding company will be listed on the portal and can be reviewed while applying. To make sure, that the candidates does not miss the deadline, system generates automated reminders and notifications for candidates.

During this process, to establish a communication between administrator and candidate, system will implement a chat module with advance LLM based summarization capabilities. To resolve the issue of huge number of messages generated, summarization tool proves to be useful to address each and every query posted in the section. A web based dashboard will also be provided to Training and Placement Department for managing and accessing student data. Through this Training and Placement Department can post job opportunities, send important notices and keep a track of all student placement activities.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature survey

Effective communication between students and TNP is one of the key to resolve all the issues faced during the placement season. With the advent of messaging apps, each TNP volunteer receives hundreds of messages everyday from all the aspiring students. It is humanly impossible to answer each and every message or interpret each one of them. To resolve this, we implement 'Text Summarization' on all the messages received and try to deliver meaning of them through a smaller message. Growing use of LLMs, the field of text summarization has been enhanced to a large extent. Following are few of the methodologies used for Text summarization.

Mark Dredze and co-authors [1] state that with increasing networking and communications within individuals, people have started using mail service to a large extent. Number of emails generated has been increasing substantially. It is humanly impossible to read and understand each and every email received from a variety of sources. To address this concern, the author along with co-authors proposes an unsupervised learning approach to generate summary of keywords for emails using latent topic models. The goal of this technique is to provide the user the gist of email in a very concise and precise way. To achieve this, the paper proposes use of Latent Semantic Analysis (LSA) and Latent Dirichlet Allocation (LDA) for generating summary keywords. The four methodologies namely LDA-doc, LSA-doc, LDA-word, and LSA-word are employed and compared against the TF-IDF as baseline. Evaluation of these methodologies is done by using Enron email corpus open source dataset which contains around 25,000 emails from 150 users.

Mohamed Abdel Fattah et al. [5] studied automatic text summarization using a trainable summarizer based on sentence-level features like position, keyword presence, length, and centrality. Their goal was to develop a system that extracted important sentences for summaries by training ML models to weigh these features. To achieve this, the authors described four techniques: Feature Extraction, Trainable Summarizer, Genetic Algorithm (GA), Mathematical Regression (MR), and Testing and Summarization. Feature Extraction identified key features, and the Trainable Summarizer implemented two overarching models. GA used natural selection to optimize feature weights, while MR correlated sentence features with manually constructed summaries. In Testing and Summarization, sentences were ranked and selected based on scores, considering a compression ratio. Training data

consisted of 50 manually summarized religious articles, while the testing data included 100 English religious articles from the Internet archive.

Jan Ulrich et al. [2] states the need to summarize email conversations due to the growing volume of email each individual receives. This paper [2] emphasizes the use of a regression based machine learning approach for email thread summarization. The research also focuses on a novel feature - speech acts and their usefulness in email summarization. It also compares various regression based classifiers and proves how they perform better than binary based classifiers. To validate the performance, it uses 10 fold cross validation for evaluation and employs weighted recall as the evaluation metric. The author uses BC3 corpus dataset and Enron email corpus open source dataset [1] to compare performance of various regression based classifiers.

Pete Burnap et al. [3] addresses the increase in social networking amongst individuals via social media platforms such as Twitter, Threads and many more. To stay aware and up-to-date with all the real world events, each individual seeks information via these platforms. This paper proposes summarization of real world event tweets and provides gist of all the recent tweets to users. To achieve this, the author states three techniques - Temporal TF-IDF, Retweet Voting and Temporal Centroid Representation for automatic summarization of generated tweets. Using Temporal TF-IDF, the words and their importance are evaluated based upon timelines whereas in the Retweet Voting approach, consider a tweet as a representative message taking the number of retweets and their timeline into consideration. The Temporal Centroid method evaluates similarity and centrality of various tweets across different clusters. Around 75,000 tweets dataset was used for evaluating various techniques and validated based upon a variety of evaluation metrics.

Jingqing Zhang [4] along with co-authors [4] showcase their work in the domain of abstractive summarization and tailoring pre-training objectives. The recent work in the domain of extractive summarization showed great success but the sector of abstractive summarization has not been explored up to its full potential. This paper from Google AI emphasizes on pre-training large Transformer-based-encoder-decoder models on massive text corpora with advanced self-supervised objectives. It introduces a State-of-the-Art algorithm for abstractive summarization with novel pre-training objectives. It employs the use of both Gap Sentence Generation (GSG) and Masked Language Model (MLM) as pre-training objectives. PEGASUS model used [4] removes/masks important sentences from an input document and generates

all together a singular output sequence from all the remaining sentences. The author tests the model across 12 different datasets related to various domains like media , science , medical , governmental and many more. The results of the model were validated using human evaluation.

Derek Miller and co-authors [7] created a lecture summary service that used BERT for extractive summarization in hopes of identifying key sentences from lecture transcripts using machine learning techniques such as BERT for text embeddings and K-Means clustering for sentence selection. It addressed some of the inconveniences related to existing systems by providing dynamic summaries for lectures. Text preprocessing involved tokenizing the transcript and discarding useless information. BERT provided improved sentence embeddings based on the pre-last layer, while K-Means clustering selected sentences closest to the centroid for the summary. It had a RESTful API based on Python that allowed users to submit transcripts and specify the preferred summary length. The authors did not name a specific dataset but mentioned that the service was tested on lecture transcripts, possibly from Udacity.

Rahim Khan and co-authors [9] developed clustering methods for extractive text summarization, particularly K-means clustering and TF-IDF. They aimed at reducing the workload of large texts and extracting key sentences for use in automatic summarization. Basic methods were included, preprocessing including tokenization and stemming. The Elbow method and Silhouette method were used to determine the best numbers of clusters (K values), grouping with respect to the importance of sentences based on the scores from TF-IDF, and their performance has been evaluated in terms of precision metrics against other summarization approaches. experimented on news articles demonstrated improved summary accuracy and readability.

Mingxi Zhang and co-authors [6] studied the performance of TextRank for keyword extraction, experimenting with parameters like co-occurrence window size, iteration count, decay factor, and text length. Their goal was to optimize TextRank for tasks such as text classification, information retrieval, and summarization. Text preprocessing involved removing stop words using the XPO6 list and applying Porter Stemmer for word stemming. The TextRank model treated keywords as nodes and co-occurrences as edges, using PageRank to rank their importance. Experiments focused on evaluating window size, iterations, and performance using Precision, Recall, and Accuracy. The authors used the Hulth2003 dataset with manually assigned keywords and the Krapivin 2009 dataset of 2304 computer science papers.

Ángel Hernández-Castañeda and co-authors [10] present an improved approach to automatic text summarization (ATS) by leveraging lexical and semantic features for keyword detection. The method combines clustering and topic modeling to enhance both the coverage and precision of extracted summaries. To identify main topic the document approach is to use sentence clustering, supported by a genetic algorithm (GA). It uses Latent Dirichlet Allocation (LDA) to model topics and determine the importance of sentences within each cluster. Then to convert text into numeric vectors authors use TF-IDF, one-hot encoding, and Doc2Vec transformation. Quality of cluster assessed using Silhouette index, and sentences selected by clusters based on the occurrence probability. Authors used 2 widely used datasets in text summarization, DUC02 dataset and TAC11 dataset, which contains multi-domain and multi-language texts and collection of news articles with human-generated summaries, which serve as the basis for performance comparison.

Zhang and co-authors [8] conducted a study on long-dialogue summarization, addressing the limitations of current transformer-based models in handling lengthy dialogues. The authors experimented with Long Former models, focusing on a retrieve-then-summarize pipeline and hierarchical dialogue encoding models like HMNet, to improve summarization performance in tasks such as meeting and media interview summarization. Preprocessing involved tokenization and truncation of dialogues when input limits were exceeded. In the summarization task, utterances were treated as nodes, with edges representing connections between two utterances, and importance was ranked using various retrieval methods, such as TF-IDF and BM25. Performance was measured using ROUGE. The datasets used included QMSum, MediaSum, and SummScreen, and the experiments highlighted the impact of retrieval and encoding methods on summarization quality.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

1. Assumptions:

- The candidate will not share his/her own credentials with anyone else.
- The candidate will act ethically and take responsibility for his/her action at every step.
- Candidate should keep an eye on Training and Placement portal for recent updates.
- Admin credentials are only kept with Training and Placement officer.
- Timely updation of job listings by TNP
- Addressing student issues by TNP through messaging system.
- No unfair means involved in data verification on admin side.

2. Dependencies:

- The candidate must possess an up-to-date desktop.
- TNP Department must have an up-to-date desktop for accessing the admin dashboard.
- Stable internet connection.

3.2 Functional Requirements

The following are the functional requirements of our Training and Placement portal

3.2.1 Registration & Verification of a candidate

A three-way authentication mechanism must be available while creating a candidate's profile on the portal. While creating profile, candidate must upload all the necessary details and submit it via the portal. Admin will then verify and approve the candidate details, and then only candidate must be able to access the hiring portal.

3.2.2 Job Opportunities Display

A sophisticated user interface for the system should be available to present a list of job opportunities. Relevant details about each opportunity such as shortlisting criteria, package offered , company and position details must be showcased in appropriate manner.

3.2.3 Personalized Notifications

The system to deliver timely reminders regarding form deadlines , online assessments and interview schedules for each and every company. Relevant notices regarding upcoming companies or any events must be shown in a concise manner.

3.2.4 Application based on Shortlisting criteria

The portal should allow candidate to apply for the job only if the he/she meets the shortlisting criteria stated by the company. A candidate not fulfilling the criteria should not be able to apply for the company.

3.2.5 Messaging module

In order to ensure a proper communication between both the parties involved during hiring process system should have an integrated messaging system. Text summarization capabilities must be integrated to enhance communication.

3.2.6 Listing Job opportunities

The portal should have a simple job listing interface through which TNP Department can list job opportunities of registered companies. Various details along with summarized job description should be listed through this module.

3.2.7 Graphical Dashboard

The system must provide a visually easy to use dashboard for administrators so as to manage and view student and registered company data. It must also provide analytical feature such as number of students placed, number of companies visited , candidates hired by each company and many more.

3.3 External Interface Requirements

The proposed system's interfaces will be divided into two main sections – the client side for end-users to interact with the system, and the admin side for managing and controlling system operations.

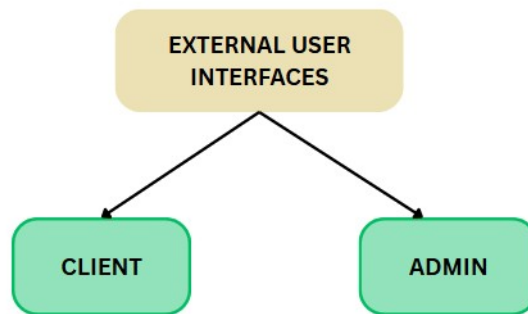


Figure 3.1: External Interface Requirements

3.3.1 Web-Based Interface - Client

The web-based interface for the client side allows students to interact with the portal. This interface enables students to register, view, and apply for job opportunities listed on the portal. The design is responsive and ensures compatibility with all major web browsers. Vue.js is used for the frontend development, ensuring a smooth and user-friendly experience. The interface will include features such as user authentication, job listings, and an easy-to-navigate application form for job applications.

3.3.2 Web-Based Interface - Administrator

The web-based dashboard for the admin side is designed to be compatible with all major browsers such as Google Chrome and Firefox. Administrators can use this interface to manage and oversee the portal. It allows admins to maintain records of student data, track the number of students placed, monitor job opportunities, and manage other system functionalities. The dashboard is implemented using Vue.js, providing a responsive and intuitive interface for efficient admin operations, such as access control, data visualization, and user management.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

1. User satisfaction: It evaluates if the application and services offered meet or exceed expectations of candidates and all stakeholders involved. The system need to be easy to use.
2. Average response time: The time it takes the Application Server to respond to a user's request and return the results is known as the average response time. Less than 10 seconds per transaction should be the system's average response time.
3. Reliability: It depends on the trust the system is able to build amongst all its users and admin.
4. Application Availability: The degree to which an application is operational, functional, and usable for completing or satisfying a user's or business's requirements is referred to as its availability.

3.4.2 Safety Requirements & Security Requirements

1. User Account Security: Account address and information related to it is well secured in the database.
2. Password integrity: Login credentials should not be shared by candidate or admin.
3. Cloud Security: AWS builds security into the core of our cloud infrastructure, and offers foundational services to help organizations meet their unique security requirements in the cloud.

3.5 System Requirements

3.5.1 Database Requirements

1. PostgreSQL & Cloudinary

3.5.2 Software Requirements

1. Backend APIs: Nodejs, Expressjs

2. Large Language Models : Gemini
3. AI Middleware : Langchain
4. Frontend: Vue JS , Vuetify
5. Containerization: Docker
6. Deployment : Microsoft Azure

3.5.3 Hardware Requirements

1. Minimum 4 GB RAM Desktop for using the website.
2. **For Cloud Deployment** - For a project serving 1500 students, deploy the backend with 2 vCPUs, 4–8GB RAM, and 20–40GB SSD storage using Azure App Service (Plan B2 or P1v2) or a B2s VM. For the database, use Azure Database for PostgreSQL with 2 vCores, 4–8GB RAM, and 50–100GB storage. Choose the General Purpose tier and enable autoscaling, automated backups, and geo-redundancy to ensure reliable performance, data safety, and scalability under peak loads.

3.6 Analysis Models: SDLC Model

The Agile methodology is a popular project management approach that emphasizes collaboration and iterative development. It allows for continuous improvement throughout the project's life cycle by enabling quick and responsive changes. The development phase of our project will follow the Agile methodology, taking advantage of its flexibility, adaptability, and customer-centric approach.

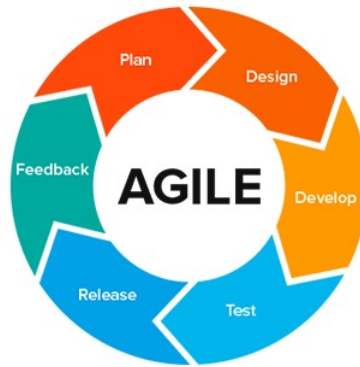


Figure 3.2: Agile Model for SDLC

In our project report, the Agile methodology will be applied to the development of the comprehensive hiring platform. We will utilize sprints, which are time-boxed iterations, to break down the problem statement into smaller modules and develop each module incrementally. Each sprint will consist of several phases, including:

1. **Plan:** During this phase, we will gather and document the specific requirements for the features and functionalities of the hiring platform.
2. **Design:** In the design phase, we will create the architectural and user interface designs for the platform based on the gathered requirements. This will involve creating use-case, UML, and system diagrams.
3. **Develop:** The development phase will involve coding and implementing the features and functionalities of the application. We will utilize the comprehensive code editor, tools, and output user interface (UI) to effectively develop the platform.
4. **Test:** In the testing phase, we will thoroughly test the developed features to ensure their functionality, performance, and reliability. This will involve both manual and automated testing approaches to identify and fix any bugs or issues.
5. **Release:** Once a feature is tested and deemed ready for production, we will deploy it to the live environment. This will involve configuring servers, setting up databases, and ensuring proper integration with other components of the platform.

6. **Feedback:** At the end of each sprint, we will conduct a feedback of the completed feature. This review will involve evaluating the implemented functionality, gathering feedback from stakeholders, and making any necessary adjustments or improvements.

By following the Agile methodology in our project, we aim to maximize collaboration, adaptability, and responsiveness, ultimately leading to the successful development of a streamlined hiring platform. The iterative nature of Agile allows us to incorporate feedback and make continuous improvements, ensuring that the final product meets the model's expectations.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

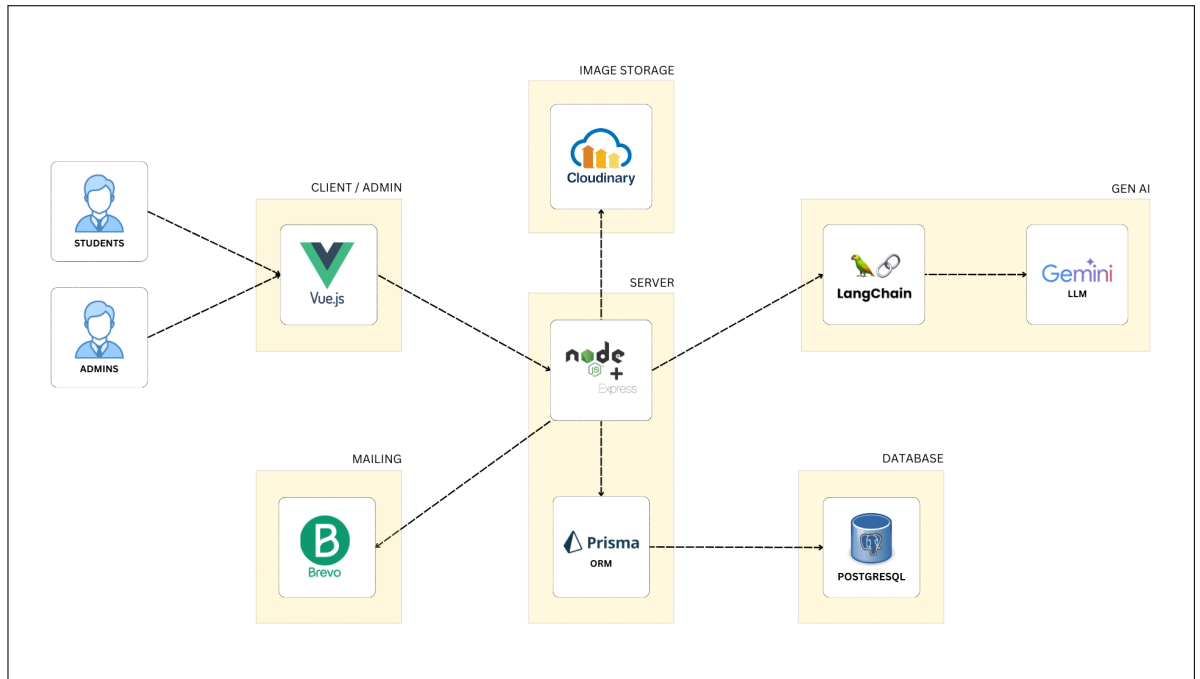


Figure 4.1: System Architecture

The System Architecture Diagram for the Training and Placement Portal involves various components.

1. **Users** : The users can interact with the system using Website.
2. **Client-side** : The cross-platform web application is built using Vue.js for users to interact with the system.
3. **Admin-side** : The admin side is designed using Vue.js to allow the admin to manage and control the system effectively.
4. **Server** : The server is created using Node.js and Express.js which handles API requests from the App and Dashboard.
5. **ORM** : The Prisma is an ORM tool used by server to interact with PostgreSQL.

6. **Database** : The PostgreSQL is used for database which is a relational database. The Prisma interacts with the PostgreSQL to retrieve and manage data as requested by the server.
7. **Cloudinary**: The Cloudinary is used for storing images on cloud. The images can be uploaded and retrieved by the server as per the need.
8. **Langchain** : The Langchain is used for processing and summarizing the text data sent by the server with the help of Gemini.
9. **LLM**: The Gemini is a LLM (Large Language Model) which is used for generating the text data.

4.2 DataFlow Diagram

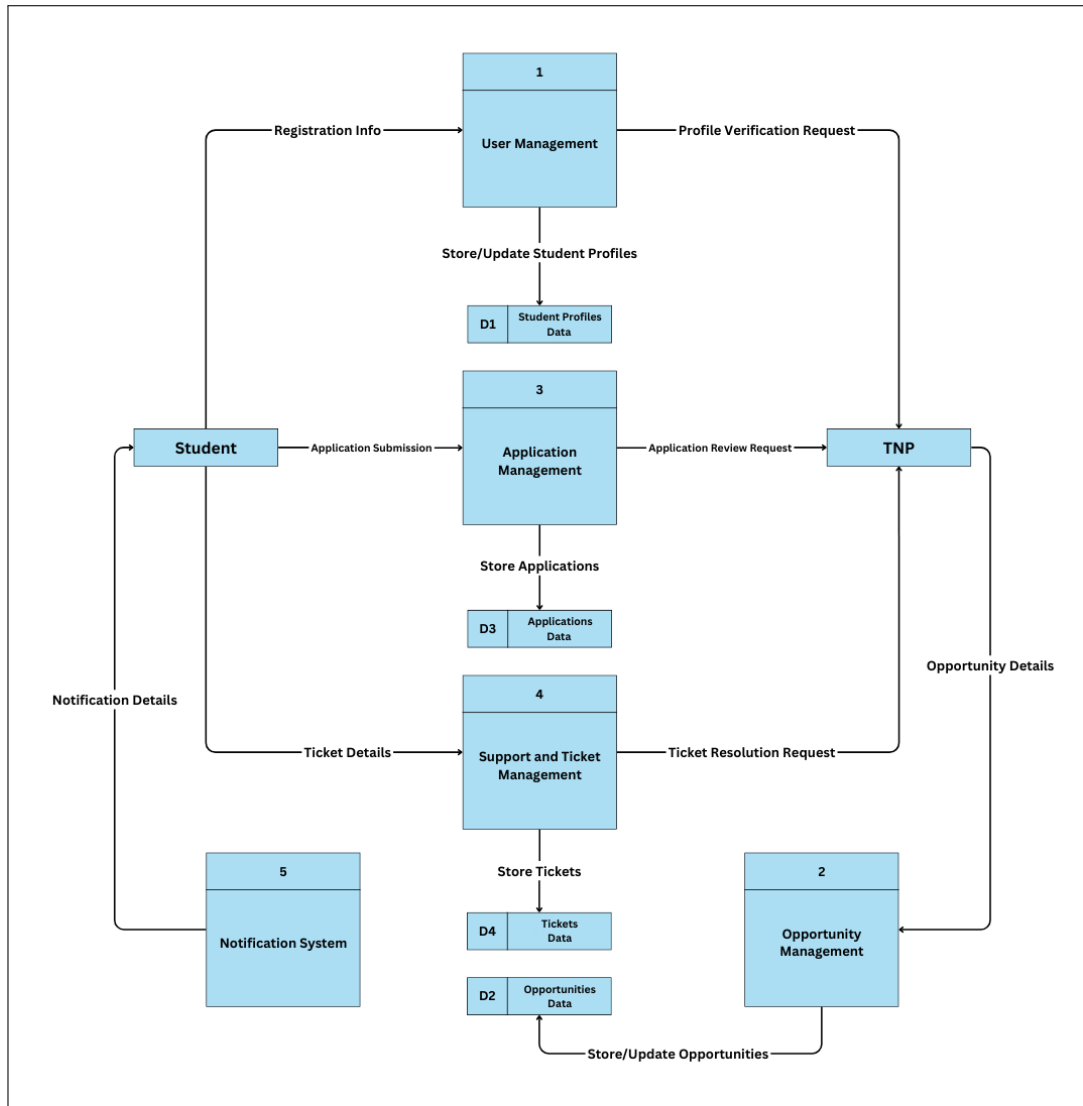


Figure 4.2: DataFlow Diagram

4.3 UML Diagrams

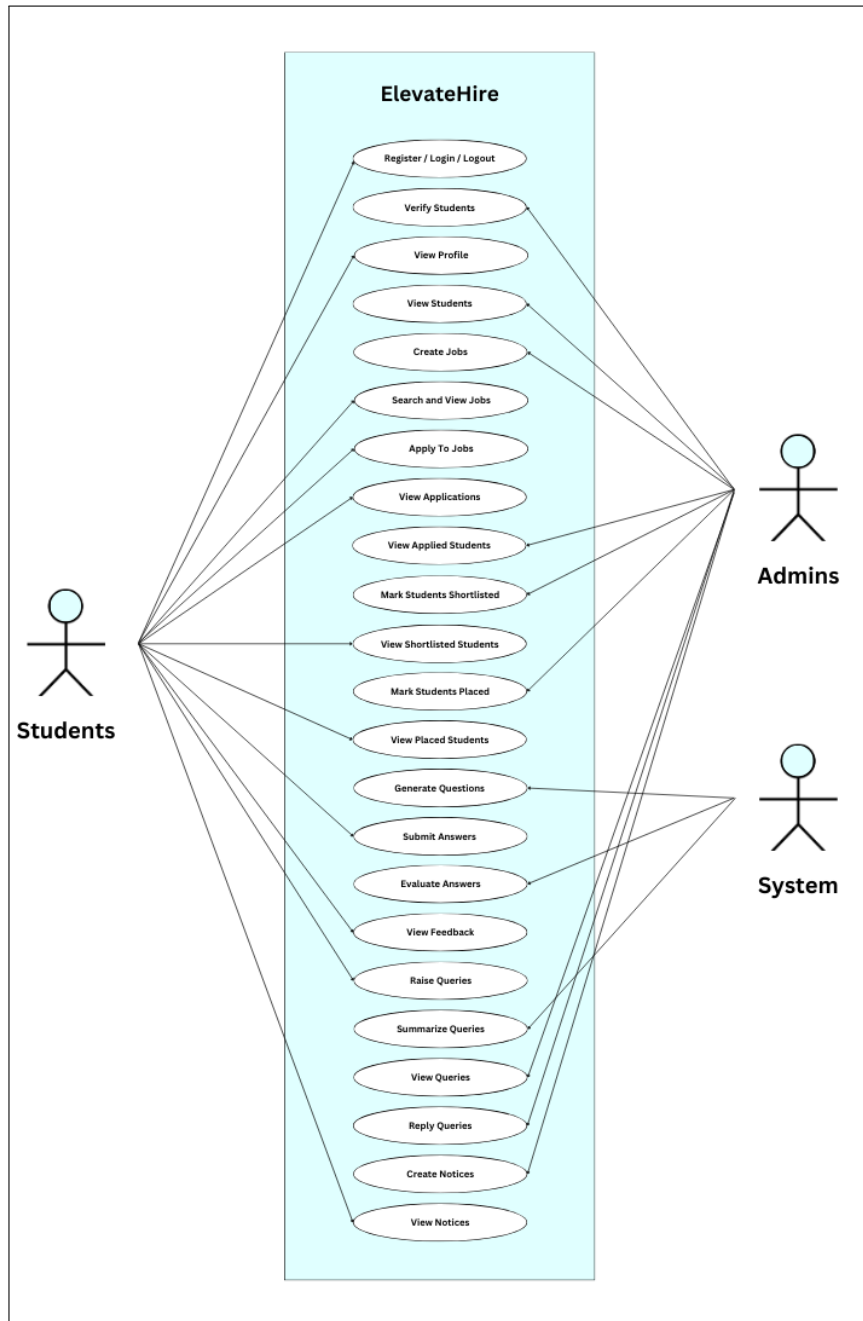


Figure 4.3: Use case diagram

4.4 ER Diagram

Here, we have shown how two different entities will interact using the system architecture.

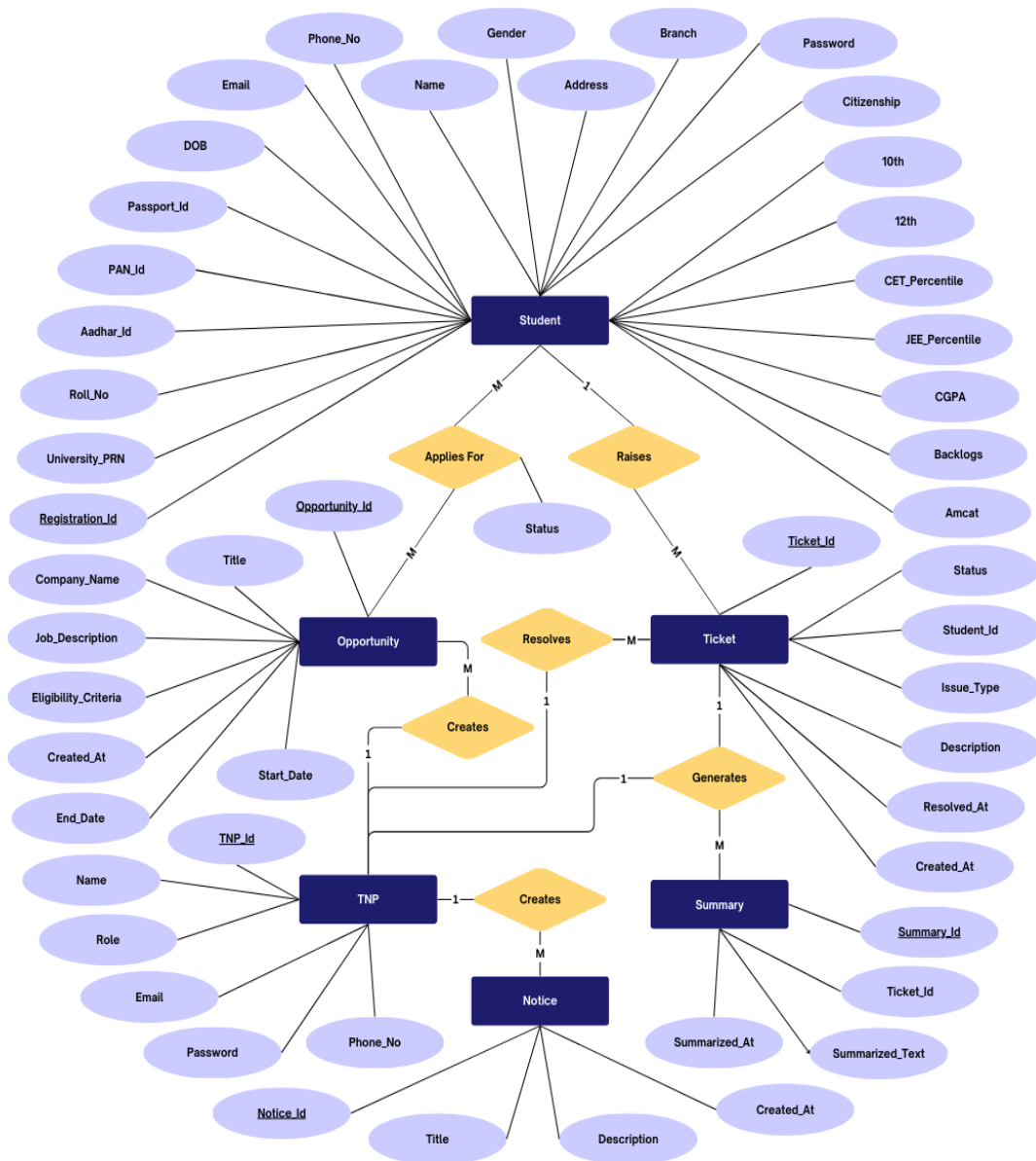


Figure 4.4: ER Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- **Research and Development:** Estimated a total of 6 months for developing and refining the core functionalities of the TNP Portal, such as student profile management, company dashboards, and automated shortlisting. This includes the time for researching existing placement systems, identifying user needs, and integrating new functionalities.
- **Testing and Evaluation:** Estimated 2 months for conducting thorough system testing, including performance testing, security checks, and user acceptance testing to ensure a seamless experience for both students and recruiters.
- **Documentation and Review:** Estimated 1.5 months to create comprehensive project documentation, including user guides, technical documentation, and system requirements. Regular review sessions will be held with mentors to gather feedback and ensure the system meets all requirements.

5.1.2 Project Resources

1. **Frontend Developers:** Responsible for designing and developing the user interface and experience using technologies such as HTML, CSS, Vuetify, JavaScript, and Vue JS.
2. **Backend Developers:** Responsible for coding and managing the backend infrastructure using technologies like Node.js, Prisma, and PostgreSQL.
3. **MLOps:** Responsible for integrating language models and machine learning algorithms into the platform.
4. **Database Technologies:** PostgreSQL and Prisma will be used for database management, ensuring efficient data storage, retrieval, and querying.
5. **Collaboration Tools:** GitHub for version control and Google Meet for team communication.

5.2 Risk Management

Risk management is an essential aspect of our project to ensure its successful implementation. By proactively identifying and addressing potential risks, we can minimize their impact on the development and deployment of our application.

By proactively identifying these risks and implementing appropriate risk mitigation strategies, we aim to minimize the potential impact on the development, functionality, and security of our application. This proactive approach will contribute to smoother project execution and enhance the project's overall reliability and performance.

5.2.1 Risk Identification

During the risk identification process for our TNP portal project, we have identified the following risks:

1. Technical Risks:

- Integration challenges with existing campus systems.
- Database performance issues leading to slow query responses during peak usage.
- Dependency on a web browser and internet connectivity for accessing the portal.

2. Business Risks:

- Budget constraints impacting feature development.
- Missed deadlines affecting the overall project timeline and launch.
- Change of requirements and issues resulting in project delays.

3. Resource Risks:

- Minimal team members with expertise in necessary technologies.
- High workload leading to team burnout and fatigue.
- Tight deadlines causing team stress.

4. Security Risks:

- Unauthorized access to the platform, leading to potential misuse of data or functionalities and data breaches.
- Sharing of candidate credentials to unauthorized persons.
- Breach of admin credentials.

5.2.2 Risk Analysis

After identifying the risks, we have conducted a risk analysis to determine their potential impact and develop mitigation strategies:

1. Technical Risks:

- Mitigation: Thorough testing and ensuring seamless integration with legacy system. Backup plans in place to address issues promptly.

2. Business Risks:

- Mitigation: Clear project goals, timelines, and budget. Regular communication with mentors to address changing requirements.

3. Resource Risks:

- Mitigation: Identifying skills gaps and taking actions like training team members as needed.

4. Security Risks:

- Mitigation: Implementing best practices for data security, regular monitoring for vulnerabilities, and conducting security audits.

5.3 Project Schedule

5.3.1 Project Task Set

- Conduct a survey to understand issues faced by students during the hiring process.
- Arranging a meet with Training and Placement office alongwith volunteers to understand their requirements.
- A formal discussion with higher college authorities to understand their expectations from the project.
- Brainstorm the idea according to the specified requirements.
- Maintaining of proper documentation stating the problem statement and concerns stated.
- Identifying the technology stack by checking in the feasibility based on various factors.
- Research innovative approaches and advance technologies to solve some prevalent issues.
- Assigning module to each team member based upon his/her expertise.
- Beginning of learning of required techstack for development of the portal.
- Designing of interactive user interfaces for student and admin side.
- Finalizing on approval the design for the complete portal.
- Development of user interfaces and functional modules simultaneously.
- Integration of frontend and backend and .
- Deployment of the project and assessing its correctness based upon opinions from higher college opportunities.
- Iteratively finding out advancements that can be made within the project.
- Testing and assessing reviews from students on the project.
- Document the entire process and code base of the project.
- Submission of project and documentation for later maintenance.

5.3.2 Timeline Chart

This section represents summarized project timeline chart. It briefly explains the schedule of our entire project from inception to projected completion.

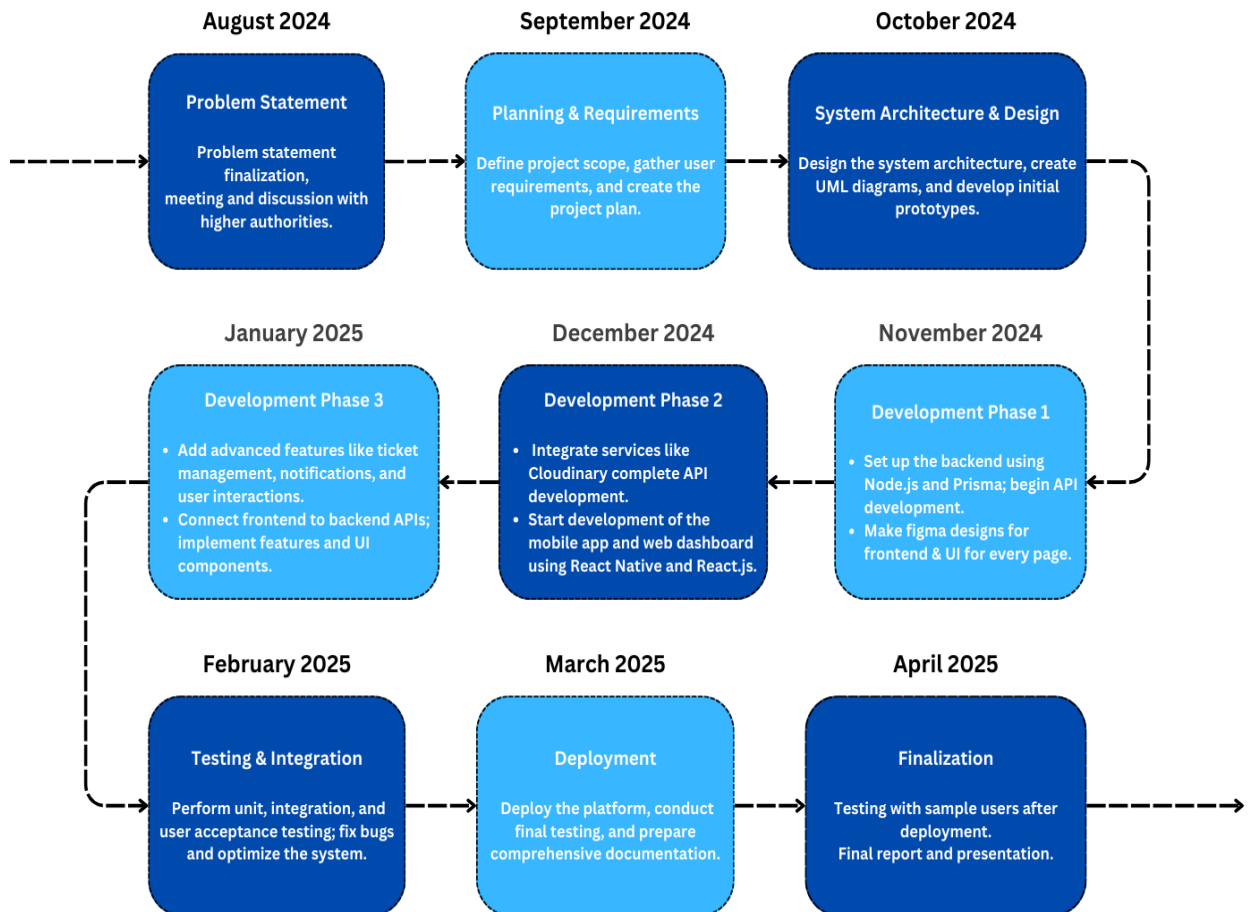


Figure 5.1: Summarized Timeline Chart

5.4 Team Organization

5.4.1 Team structure

- **Atharva Litake:** Frontend Developer, UI/UX Designer, Leader.

- **Kishanlal Choudhary:** Backend Developer, System Designer.
- **Apurva Kulkarni:** Frontend Developer, Documentarian.
- **Aditya Darade:** Database Engineer, Backend Developer.
- **Prof. Pranali R. Naoghare:** Internal mentor and guide.
- **Dr. Geetanjali V. Kale (HOD CE):** Internal mentor and co-guide.
- **Dr. Prahlad T. Kulkarni (Director, PICT):** Advisor and higher leadership.

5.4.2 Management reporting and communication

- **Management:** For management during our project, we used several effective tools and practices to ensure regular and timely updates. We used Google Groups for team communication, where we regularly shared updates, research papers and important documents. Additionally we collected various research papers related to our project and organized as well as structured them properly within Google Drive. This made it easy for the entire team to access and refer the collected material. For code management and version control, we used Git and GitHub, ensuring that all team members can collaborate, track changes and contribute to project.
- **Reporting:** Bi-weekly, we reported our progress to the internal mentor, and provided updates on work completed, challenges faced and ideas we came up with. The regular communication allowed us to stay aligned with project goals and deadlines of submissions. Furthermore, we continuously updated our co-guide Dr. G. V. Kale with new development and insights, seeking valuable feedback. Based on this, we consulted our advisor Dr. P. T. Kulkarni for new ideas and strategic guidance, which we will be implementing in our project, ensuring improvements and innovation.
- **Communication:** For regular communication in team members we used Google Groups and Whatsapp for quick updates, making it easy to share new information and get immediate feedback. Our communication with the mentor was structured around bi-weekly sessions and regular updates and tasks were given accordingly.

Thus, this structured approach to management reporting and communication helped us to maintain clarity, accountability and efficiency throughout the project.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

6.1.1 Client Side

1. **Login Module** - This module is responsible for handling user credentials such as registration number and password, validate it, and generate a session/JWT token if the user is authenticated. This module ensures that the website is protected from unauthorized users entering the system.
2. **Job Listing Module** - The website aims to provide students the opportunity to see companies that are hiring through the college. Once the company is on board then there will be information in a job section on the website with company name, role and date of visit as the basic details. Each job will have a view details button, which when clicked would show any other details such as: eligibility, CTC, skills needed for the job and a job description. In this "view detail" if the student clicks on the apply for job button, students can apply for that job right then. Overall, yes it does make it easier for the student to apply for jobs, and help students to keep track of placement jobs they are applying for.
3. **Track Applications Module** - The Application Tracker module allows students to view job applications they submitted through the portal. The Application Tracker provides students with information about the company, job title and application status (e.g. Applied, Shortlisted, Selected). Students can view their status through all of the various steps of the job application process and receive any live updates that may occur during the job application process. This feature provides a centralized view of all applications, helping students stay organized and informed throughout the placement process with improved transparency and ease.
4. **Results Module** - The system displays a list of placed and shortlisted students per company that can be accessed by both the TNP Cell and the students. Once a student has been marked placed by a company, that student will no longer be able to apply for other companies. This is to ensure fair opportunities to others and to encourage more honesty in the placement's process.
5. **Student Profile Module** - Students have access to their profile, where they can find important information, such as personal information, aca-

demic record, and placement status. There is also a dynamic line graph to visually display their CGPA each semester, which provides a good means to visualize their performance and their grading before and after that semester. Overall, a succinct opportunity to graphically represent their educational journey with a small organization and clarity.

6. **Help and Support Module** - Students can directly send query to the TNP Cell about their concerns through this section. This increases communication speed and guarantees answers to student issues promptly and to ensure a smooth and transparent relationship between the student and the placement team.
7. **Notices Module** - Students receive global notices for important updates like form deadlines, company info, and interviews. This keeps them informed and avoids missed opportunities.
8. **PrepAI Module** - The system provides students with assessments, assesses their comprehension of material, assigns ratings and a final score. Depending on their performance, AI supplies personalized suggestions for improvement to develop students' weak areas and increase their readiness for placement.

6.1.2 Administrator Side

1. **Login Module** - This module handles the user authentication part of an application (logins, emails, password, etc.), checks for validity, and creates a session/JWT token if the user is authorized. This guarantees that the website will be secure from unauthorized users of the system. Also, the system has a login method that is secure ensuring privacy and access to admins.
2. **Dashboard Module** - The dashboard allows the administrator to view the placement status of students placed or unplaced, with a breakdown of the information by department to assist in management of decision making.
3. **Add Job Opportunity Module** - The administrator has the ability to easily post job opportunities with all relevant company details, job descriptions, full package offered, eligibility requirements, and application deadlines, which will guarantee that students receive accurate job advertisements in a timely manner.

4. **Verify Students Module** - The administrator has the ability to view and review the personal data of all students from the registration process, to ensure the information that is presented is accurate. This keeps the data up to date and ensures consistency throughout the system.
5. **Mark As Placed and Shortlisted Module** - The administrator can change a student's status to "Placed" when the company has selected the student or can change a student's status to "Shortlisted" if the student passed the online assessment to ensure proper and accurate tracking of the recruiting process.
6. **Export Student Data Module** - The administrator can export student information into a CSV file for easy downloads and sharing with companies. This makes sharing student data for recruitment a streamlined process, while maintaining appropriate records.
7. **Send Notices Module** - Using a single notification module, the system allows the admin to notify users about important updates, providing the opportunity for administrators to notify students via timely communication about school-related job opportunities, deadline announcements, and much more.
8. **Help and Support Module** - The platform is designed for effective handling of student questions, with features to summarize multiple inquiries or condensation of lengthy text to aid time-stamped review, and an option to email the students directly for verification of student engagement and support quickly as possible.

6.2 Tools and Technologies used



Figure 6.1: Tools and Technologies

6.2.1 Frontend Technologies

1. **Vue.js** - Vue.js is a javascript framework that helps developer build featuristic frontend interfaces. Vue.js is well known for its easy implementation , easy learning curve , and its ability to be intergated in legacy projects. It is also known for its modular style of writing code.
2. **Vuetify** - Vuetify is a Vue component framework used to easy design and use ready made Vue components. It is also called as no design skills framework where the developer can design the complete website with minimal amount of efforts on design side. Vuetify is also used for maintaining consistent design of web interfaces with code reuasability.
3. **Figma** - Figma is a UI/UX tool that is used to design and prototype user interfaces. It allows the designers to design the product and show-case their ideas to the developers of how the finished product should look like. It gives a blueprint of the product for developers.

6.2.2 Backend Technologies

1. **Node.js** - Node.js is an open-source , cross platform javascript runtime environment that allows execution of javascript outside the browser. It allows developers to do server side scripting and develop the API that can communicate with the client application. It has a large development community and support forums.
2. **Express.js** - Express.js is a fast, minimal Node.js web framework used to build APIs and web applications, offering powerful routing, middleware support, and easy integration with databases and templating engines. It also offer NPM packages for easier development and code reuse ability.
3. **Prisma** - Prisma is an object-relational mapper for Node.js and TypeScript application that helps easier interaction with databases. It helps developers work with databases more efficiently and reliably by abstracting away complex SQL queries and handling database schema changes.

6.2.3 Databases

1. **PostgreSQL** - PostgreSQL is an open source object relational database that offers both relational as well as object-oriented features. It has over 35 years of active development and large community for help and support. It provides various robust data protection and storage features.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

To ensure that the TNP Portal is operating properly, securely, and as intended, several types of testing are applied throughout the development life cycle. Both the Frontend as well as the Backend of the portal needs to be thoroughly tested so as to ensure lesser discrepancies while operating.

7.1.1 Frontend Testing

1. **Sanity Testing** - During the development of any software using Agile Methodology, Sanity testing becomes a very important part for ensuring proper functioning of developed modules. Sanity testing is a type of testing that ensures that the functionality of the developed module is not hampered by the development of upcoming modules. It requires the least manpower and very few test cases to make sure that the developed product is working correctly. This type of testing is a subset of regression-based testing.

Sanity Testing of TNP Portal

- With the development and integration of the ER system in the registration, the functioning of the registration page is not hampered.
 - Appropriate working of Job Application module with concurrent development of Mark as Placed and Mark as Shortlisted Module.
 - Bug free working of basic application criteria on addition of special eligibility criteria as per demand.
2. **Manual Testing** - Manual testing involves testing of the product by the tester manually without use of any Automation tools. This type of testing is usually done to discover end user interaction pattern with the software. This type of testing takes a lot of time as the test cases are executed manually step-by-step.

Manual Testing of TNP Portal:

- Registering as a student and verifying if all the filled details are visible to admin.
- Viewing job details and able to apply to job based upon the created profile.

- On clicking start test in PrepAI module, whether or not able to see the questions , answer and submit them.
 - Upon clicking Summarize Query option, whether or not the summarized text of below mentioned queries is visible.
3. **E2E Testing** -E2E testing is very similar to manual testing where the user experience is simulated for understanding limitations of the product developed. The only difference between manual and E2E testing is use of automation tools like Cypress in E2E Testing. Cypress is an javascript based automation tool that is designed to automate the testing process. Cypress is used by QA and Frontend Engineers involvd in product development life cycle.

E2E Testing - Cypress

- Login and Logout functionality: Students, administrators, and recruiters are directed to their respective dashboards after logging in successfully.
- Students are able to raise queries through the Help Section provided on the client side.
- A student is not able to apply for a job more than once, or for jobs they are not eligible to apply for.
- Make sure all students are sent an email alert once their query is resolved.

7.1.2 Backend Testing

API Testing - API Testing mainly focuses on testing the functionality, reliability and security of Application Programming Interfaces. API's play an important role in streamline functioning of the product. It also makes sure synchronized communication of Frontend and Backend of the product. All these features of API makes API Testing important. Various tools such as Swagger, Postman, and many more are used in API Testing

API Testing of TNP Portal

API Testing of TNP Portal is done using Swagger. Swagger is a tool for design , document and test API. It is an open-source tool by OpenAPI Specification.

Training and Placement Portal

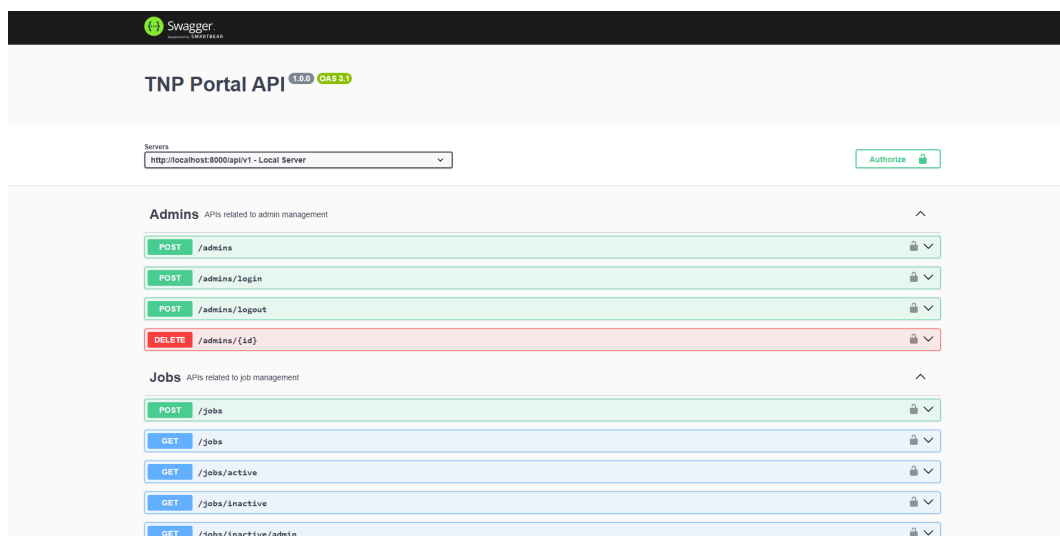


Figure 7.1: API Testing - Swagger

7.2 Test cases and Test Results

Client

Test Case	Expected Result	Actual Result	Status
Login with valid credentials	Redirect to next Page	Redirect to next Page	Pass
Login with invalid credentials	Show error, no redirect	Show error message	Pass
View student profile	Display full profile info	Info displayed correctly	Pass
Update student information	Info saved successfully	Info updated	Pass
Mark student as placed	Status = Placed	Status updated	Pass
Mark student as shortlisted	Status = Short-listed	Status updated	Pass
Reapply alert for placed student	Alert popup shown	No alert displayed	Fail
Answer support query	Mark as resolved	Resolved	Pass
Delete student profile	Record should be removed	Not removed	Fail
Apply to a job	Show Apply button + working	Working fine	Pass
Display job listings	Show available jobs	Jobs displayed	Pass
Track submitted applications	Show application statuses	Displayed correctly	Pass
Live job status update	Real-time reflection	Reflected instantly	Pass

Test Case	Expected Result	Actual Result	Status
Display result status	Status listed by company	Correctly shown	Pass
Restrict reapplication	Block second application	Reapply blocked	Pass
Show full student profile	Information + Placement Graph shown	Accurate info	Pass
Send query via Help section	Confirmation shown	Received confirmation	Pass
Send notice to all students	Notice visible to all	Displayed correctly	Pass
Reply to student query	Status=Resolved	Updated correctly	Pass
Show student-side notices	Job/deadline alerts visible	Displayed as expected	Pass
Unanswered test case (PrepAI)	No submit, no result shown	Got result incorrectly	Fail

Admin

Test Case	Expected Result	Actual Result	Status
Dashboard Module	Shows placement data with dept-wise breakdown.	Dept-wise placement data displayed.	Pass
Add Job Opportunity	Job posted with company, role, CTC, eligibility, deadline.	Job added with all details.	Pass
Mark Placed and Shortlisted	Status set to 'Placed', student can't apply to other jobs and Status set to 'Shortlisted' post-assessment.	Status updated and status updated to 'Shortlisted'	Pass
Export Data	Full student data downloaded in CSV.	CSV downloaded with correct data.	Pass
Send Notices	All students receive notice in portal.	Notice displayed to all students.	Pass
Help and Support	Admin replies, marks as 'Resolved'.	Queries handled and resolved and can reply personally	Pass
Summarize support queries	Combine queries	Summarized correctly	Pass
First time loading of jobs	Jobs should be displayed	Jobs not displayed	Fail

CHAPTER 8

RESULTS

8.1 Screenshots of website

8.1.1 Client Side

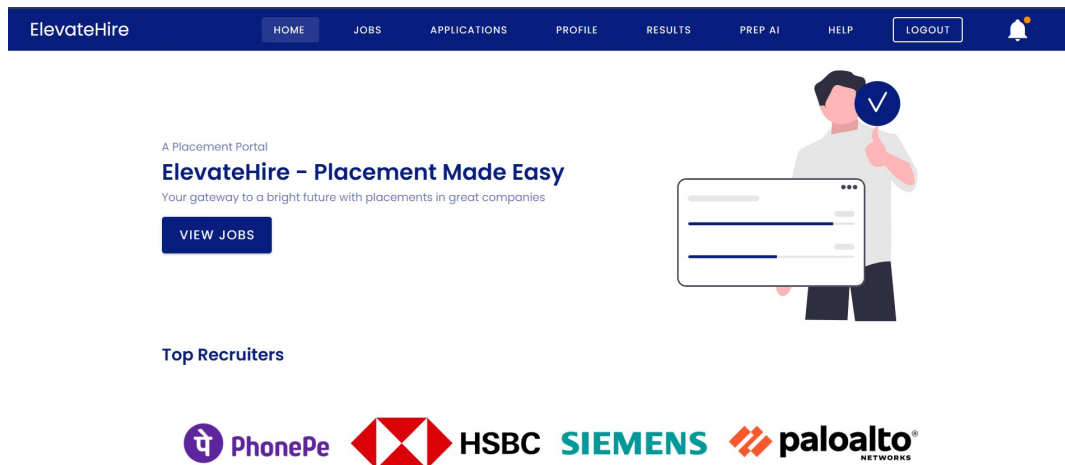


Figure 8.1: Home page

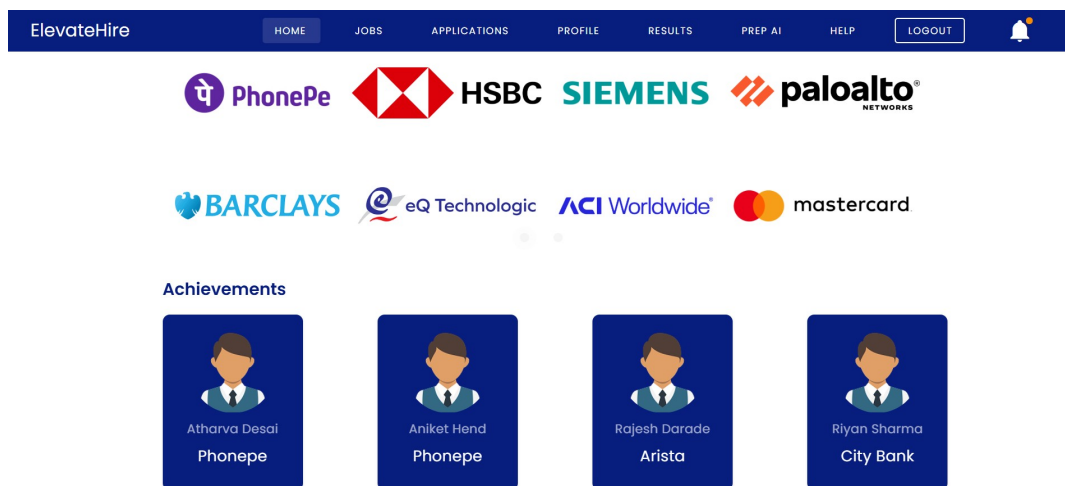


Figure 8.2: Home page

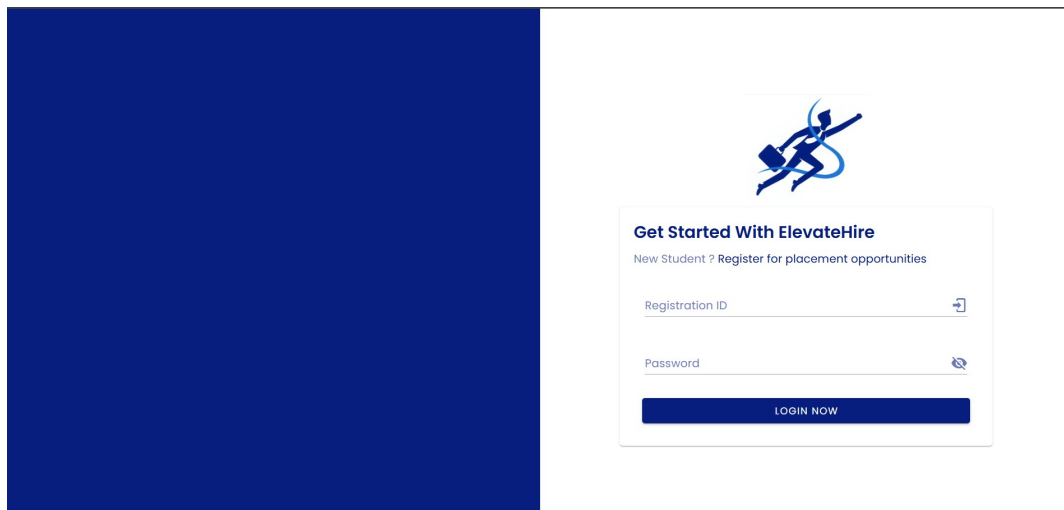


Figure 8.3: Login page

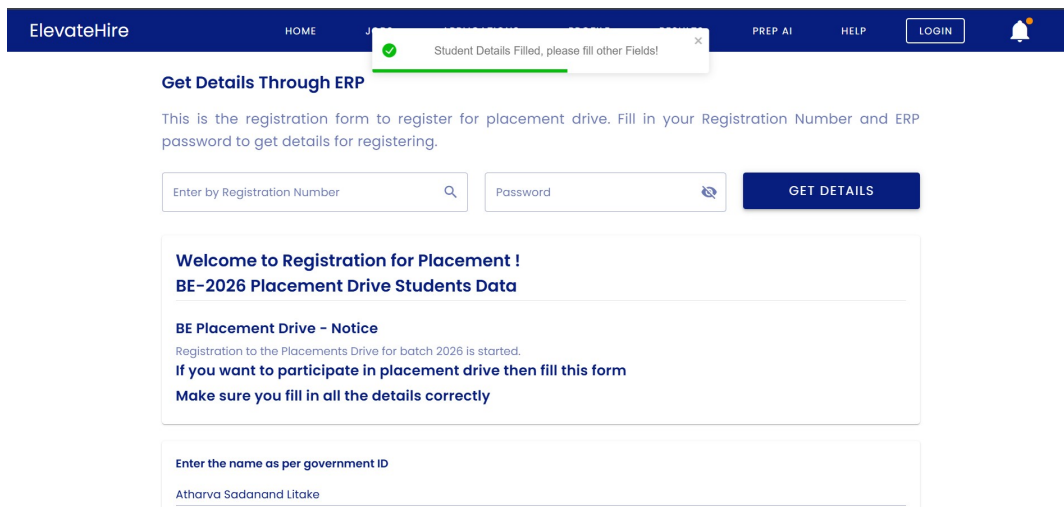


Figure 8.4: Registration page

Training and Placement Portal

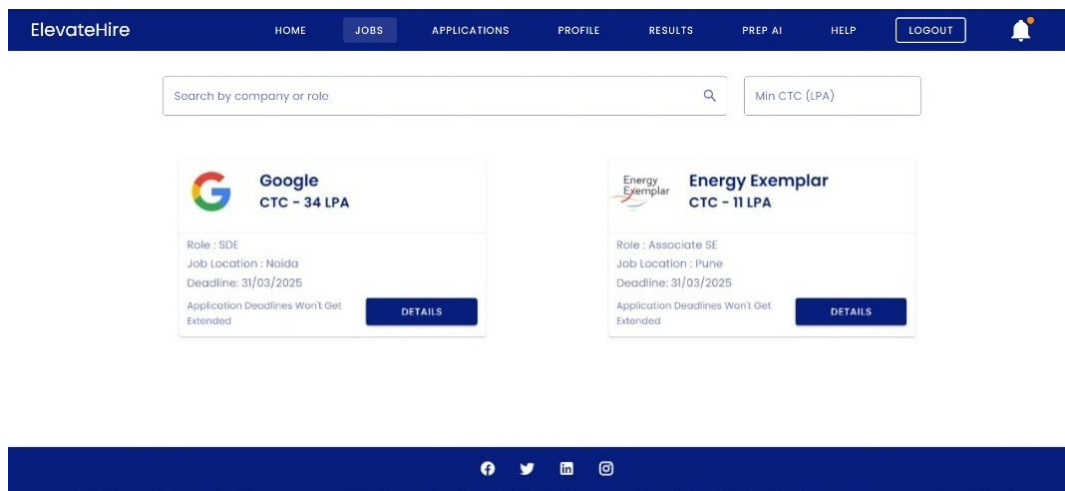


Figure 8.5: Jobs listing page

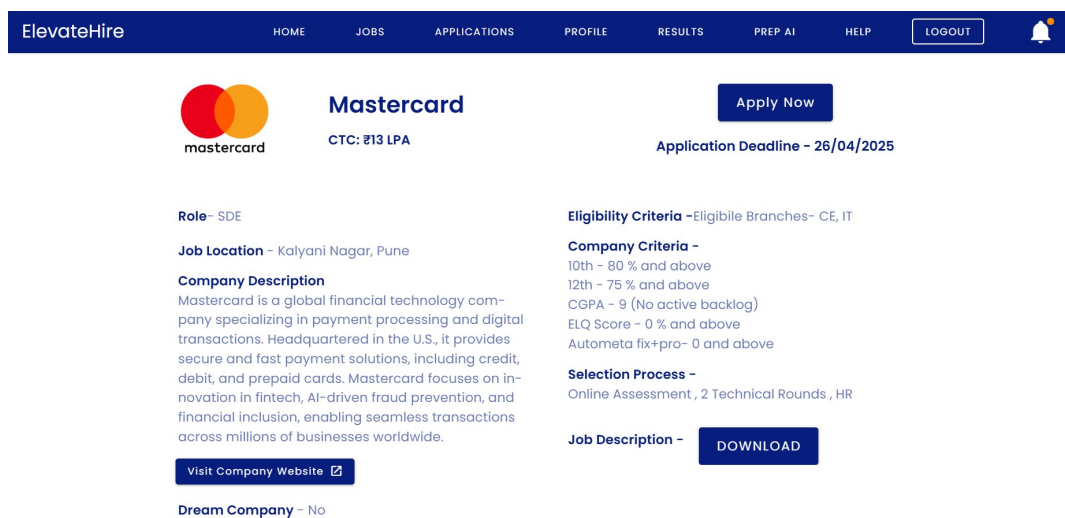


Figure 8.6: Job Details page

Training and Placement Portal

ElevateHire

HOME JOBS APPLICATIONS PROFILE RESULTS PREP AI HELP LOGOUT

Your Applications

	Barclays CTC - 14.5 LPA	APPLIED
	Energy Exemplar CTC - 11 LPA	APPLIED

ElevateHire is a smart, automated Training and Placement Portal designed to streamline the hiring process for students and the TNP Department. It serves as a one-stop platform for job listings, eligibility tracking, and seamless student-company interaction, eliminating manual errors and inefficiencies. With intelligent notifications, real-time updates, and effortless communication, ElevateHire keeps students informed and empowered—no more juggling multiple forms and deadlines.

2025 – Team – ElevateHire

Figure 8.7: Applications page

ElevateHire

HOME JOBS APPLICATIONS PROFILE RESULTS PREP AI HELP LOGOUT

Apurva Ajit Kulkarni
Engineering Branch - CE
Registration Number - C2K2221
University PRN - 72223781C

BE Roll No - 41432
Date of Birth - 2003-07-24
Gender - Female

Contact Details
Primary Email - litakeatharva24@gmail.com
Mobile No. - 7447484859

Personal Details
Aadhar Number - 669712348875
PanCard - NIOPQ5597M
PassPort - C2606712

Educational Details
10th Percentage - 90.8%
Diploma Percentage - 96.6%

Amcat Result Details
ELQ Score - 74
Automata Score - 65

SGPA Improvement Graph

Figure 8.8: Profile page

Training and Placement Portal

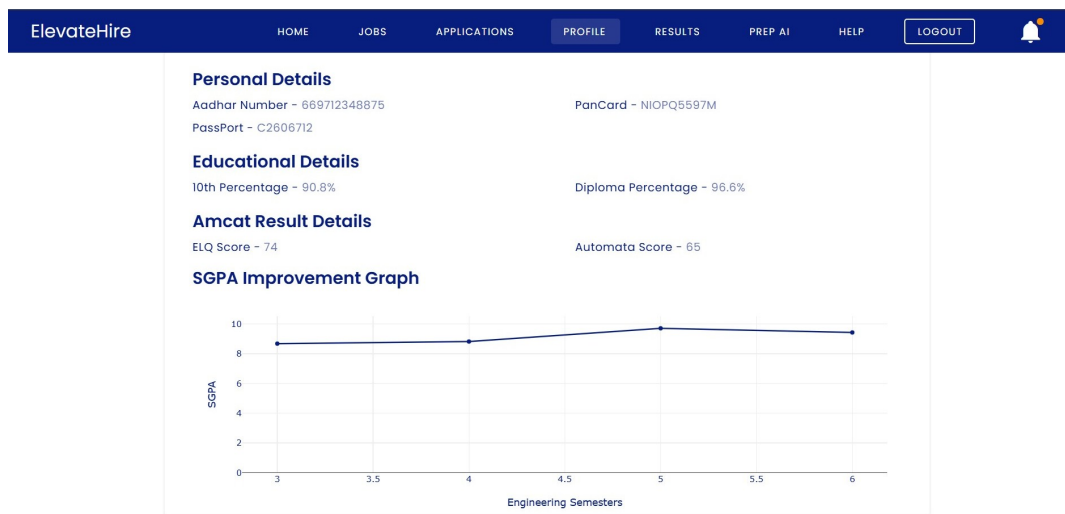


Figure 8.9: Profile page

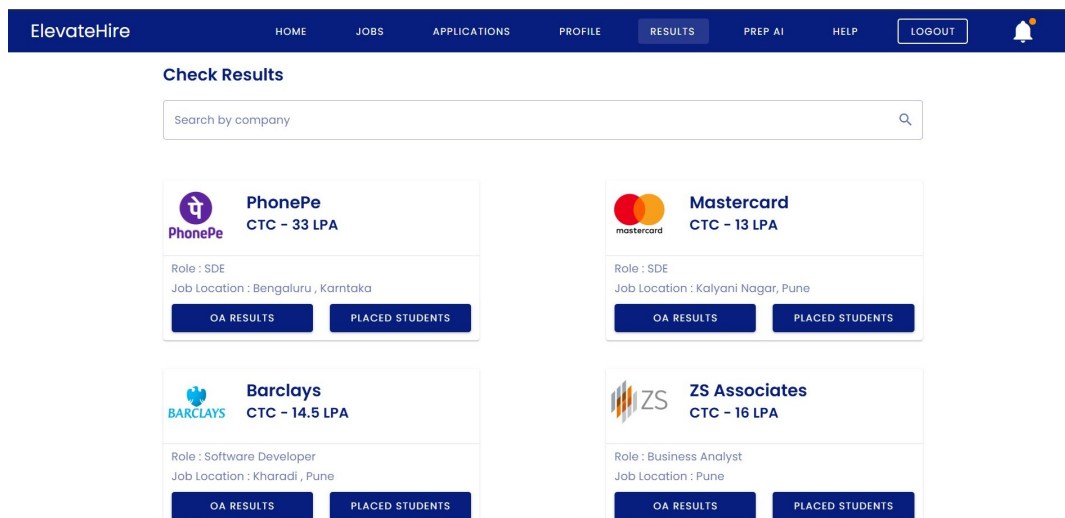


Figure 8.10: Result page

Training and Placement Portal

ElevateHire HOME JOBS APPLICATIONS PROFILE RESULTS PREP AI HELP LOGOUT

Help & Support

This section is designed for students to submit queries directly to the Administrator of ElevateHire, ensuring quick communication and efficient resolution of placement-related concerns.

Send Query

Enter your query

Maximum 100 characters allowed

SEND QUERY

Figure 8.11: Help and Support page

ElevateHire HOME JOBS APPLICATIONS PROFILE RESULTS PREP AI HELP LOGOUT

Notices

ZS Associate Registration DeadLine
ZS Associate Registration Drive will close tomorrow. Make sure you apply.

HSBC form deadline in 15 mins
HSBC form will be closed at 3pm sharp fill details ASAP.

ElevateHire is a smart, automated Training and Placement Portal designed to streamline the hiring process for students and the TNP Department. It serves as a one-stop platform for job listings, eligibility tracking, and seamless student-company interaction, eliminating manual errors and inefficiencies. With intelligent notifications, real-time updates, and effortless communication, ElevateHire keeps students informed and empowered—no more juggling multiple forms and deadlines.

2025 — Team - ElevateHire

Figure 8.12: Notices page

Training and Placement Portal

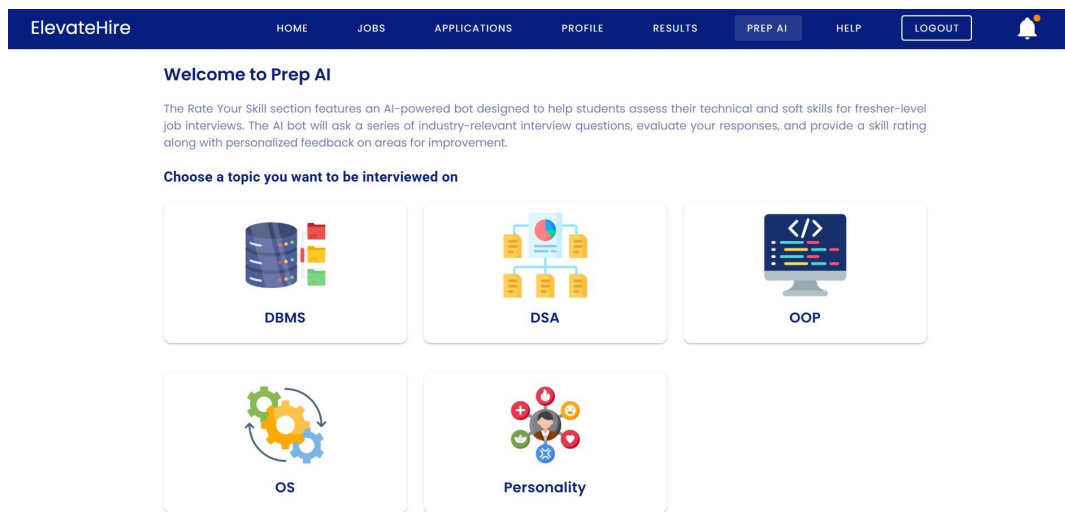


Figure 8.13: PrepAI page

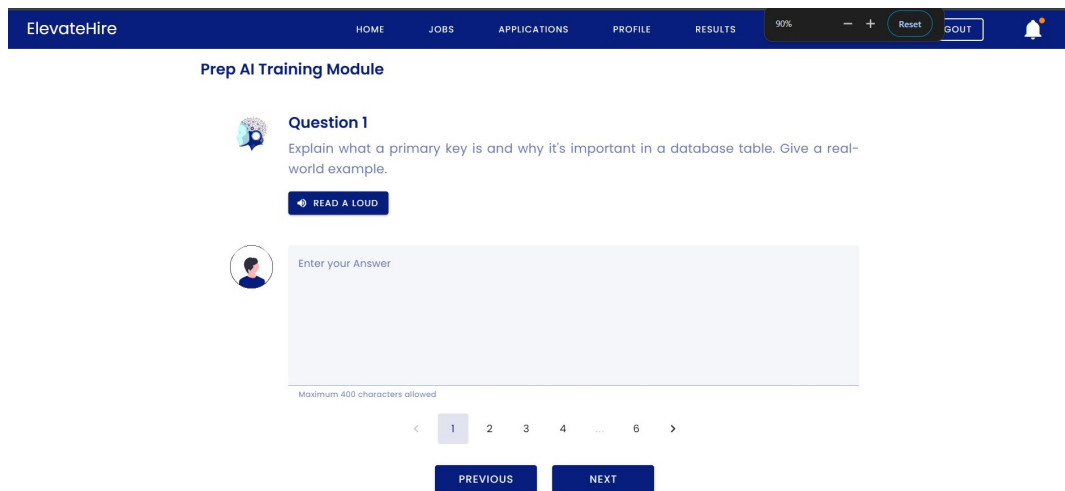


Figure 8.14: Test page

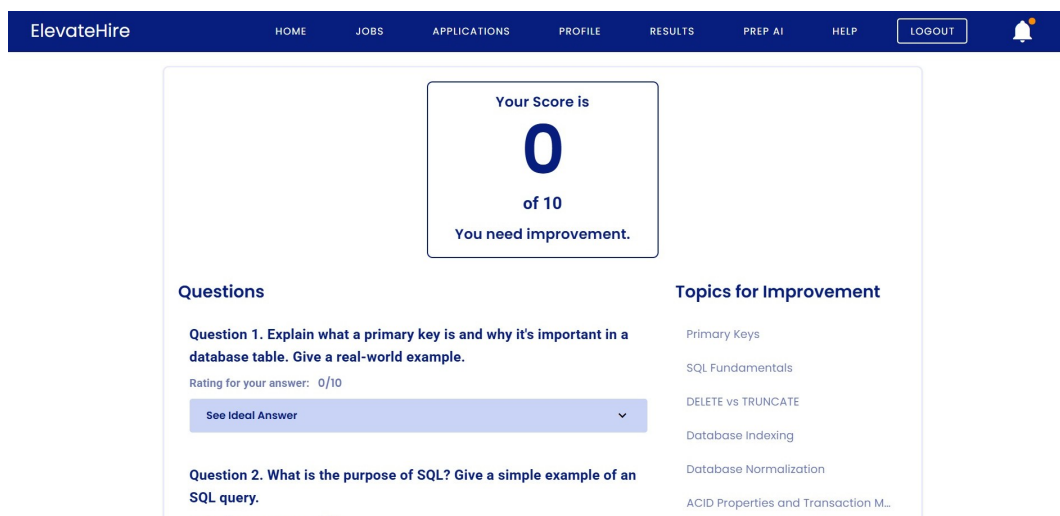


Figure 8.15: PrepAI Result page

8.1.2 Administrator Side

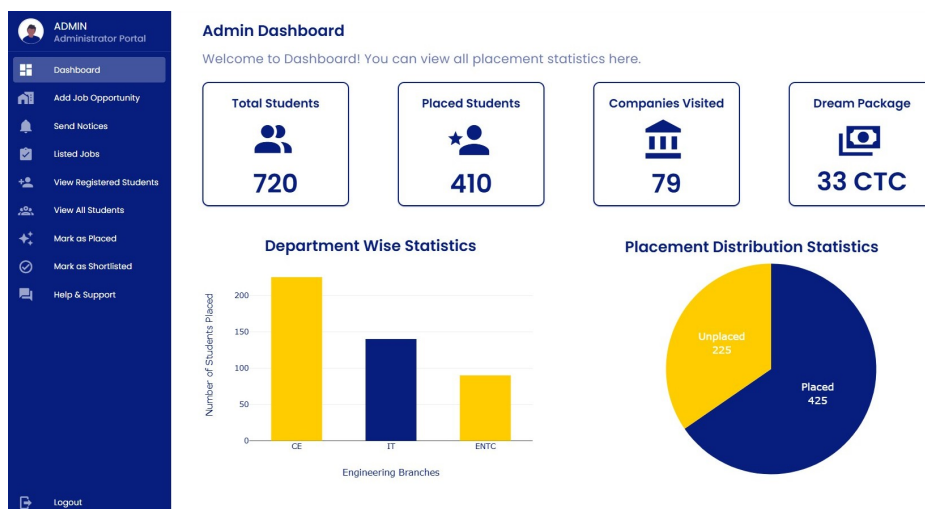


Figure 8.16: Dashboard

Add Company
Through this page, you can add details about companies that are going to conduct placement drives in the college, helping students stay informed about upcoming opportunities and recruitment processes.

Company Name

10th Percentage Criteria

Enter the Package of Company

12th Percentage Criteria

Company Logo File

Diploma Percentage Criteria

Job Role

CGPA Criteria

Job Location

Active Backlog

Company Description

Passive Backlog

Automata Score

Maximum 100 characters allowed

ELQ Score

Figure 8.17: Add Job Opportunity

Training and Placement Portal

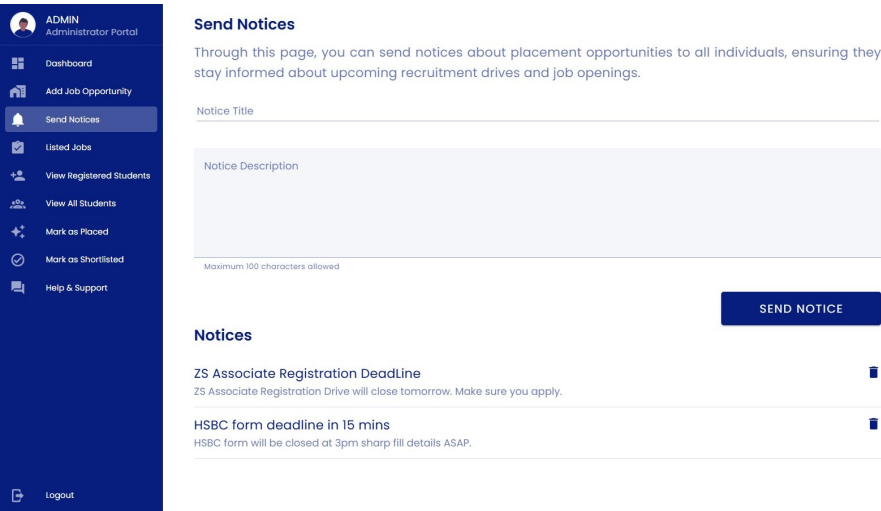


Figure 8.18: Send Notices

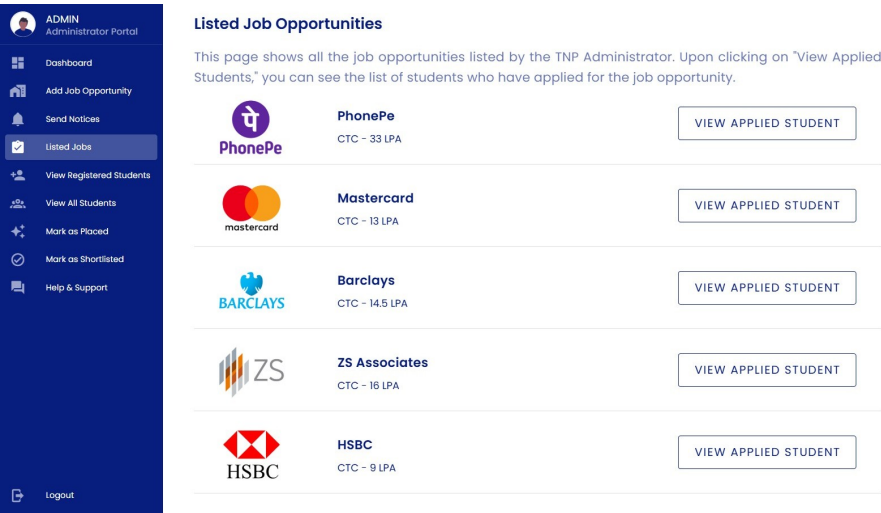


Figure 8.19: Listed jobs page

Training and Placement Portal

ADMIN
Administrator Portal

Dashboard

Add Job Opportunity

Send Notices

Listed Jobs

View Registered Students

View All Students

Mark as Placed

Mark as Shortlisted

Help & Support

Logout

View All Students

Through this page, you can see all the students registered on this portal for the placement opportunities. Click on View Details to check each and every student details.

Search by Name, Registration_ID or PRN

Student Name	Registration Number	University PRN Number	Placed	View Verified Student Details
Atharva Sadanand Litake	C2K2I106776	72278822D	Yes	DETAILS
Apurva Ajit Kulkarni	C2K222I	72223781C	No	DETAILS
Kishanil Chhelaram Choudhary	C2K2I106773	72278549G	Yes	DETAILS
Abhishek Saware	C2K2I106777	72278822A	No	DETAILS
Manas Jadhav	C2K2I106778	72278822B	Yes	DETAILS
Kamlesh Kumar	C2K2I106779	72278822C	Yes	DETAILS
Aditya Ashru Darade	C2K2I106899	722721124	No	DETAILS
Amav Kudare	C2K2I106780	72278822E	No	DETAILS
Nitin Chandra Sahu	C2K2I106781	72278822F	No	DETAILS
Shrey Sahay	C2K2I1067782	72278822G	No	DETAILS

Figure 8.20: View All Students

ADMIN
Administrator Portal

Dashboard

Add Job Opportunity

Send Notices

Listed Jobs

View Registered Students

View All Students

Mark as Placed

Mark as Shortlisted

Help & Support

Logout

Student Details

Full Name:	Apurva Ajit Kulkarni	Primary Email:	litakeatharva24@gmail.com
Alternate Email:	atharva24@gmail.com	Primary Mobile Number:	7447484859
Alternate Mobile Number:	9225582111	Date Of Birth:	24/07/2003
Gender:	Female	Current Address:	Flat no: 302, Takshashila society, Narhe
City:	Pune	State:	Maharashtra
Permanent Address:	Flat no: 302, Takshashila society, Narhe	Branch:	CE
Division:	BE-4	Roll Number:	41432
University PRN:	72223781C	Pict Registration Id:	C2K222I
Percentage10th:	90.8	Board10th:	SSC
Passing Year10th:	2019	No Of Gap Years After10th:	N/A
Reason Of Gap After10th:	N/A	After10th Appeared For:	Diploma

Figure 8.21: View Student Details

This page displays a list of students who have applied to the company. The admin can mark students as Shortlisted if they have cracked the OA, by selecting the corresponding checkboxes. An Automatic mail will get triggered to all shortlisting students.

MARK AS SHORTLISTED

Student Name	Registration Number	University PRN Number	COPA	Shortlisted	Shortlist Student
Atharva Sadanand Litake	C2K2I106776	72278822D	9.75	Yes	☐
Kishanlal Chheleram Choudhary	C2K2I106773	72278549G	8.9	Yes	☐
Abhishek Saware	C2K2I106777	72278822A	8.88	Yes	☐
Manas Jadhav	C2K2I106778	722788228	8.43	Yes	☐
Kamlesh Kumar	C2K2I106779	72278822C	8.38	Yes	☐
Aditya Ashru Darade	C2K2I106899	722721124	8.21	No	☐

Items per page: 10 1-6 of 6 < >

This page displays a list of students who have applied to the company. The admin can mark students as Placed by selecting the corresponding checkboxes to manage the hiring status effectively.

MARK AS PLACED

Student Name	Registration Number	University PRN Number	CGPA	Placed	Mark as Placed
Atharva Sadanand Litake	C2K21I06776	72278822D	9.75	Yes	☐
Kishanlal Chhelaram Choudhary	C2K21I06773	72278549G	8.9	Yes	☐
Abhishek Saware	C2K21I06777	72278822A	8.88	No	☐
Manas Jadhav	C2K21I06778	72278822B	8.43	Yes	☐
Kamlesh Kumar	C2K21I06779	72278822C	8.38	Yes	☐

Items per page: 10 1-5 of 5 |< < > >|

PICT, Department of Computer Engineering 2024-25

Training and Placement Portal

The screenshot shows the 'Help & Support' section of the Training and Placement Portal. On the left is a dark blue sidebar with a user profile 'ADMIN Administrator Portal' and a list of navigation items: Dashboard, Add Job Opportunity, Send Notices, Listed Jobs, View Registered Students, View All Students, Mark as Placed, Mark as Shortlisted, and Help & Support (which is highlighted). At the bottom of the sidebar is a 'Logout' button. The main content area has a white background. It starts with the 'Help & Support' title, followed by a paragraph explaining its purpose for resolving student queries. A 'SUMMARIZE QUERIES' button is on the right. Below this is a 'Summarized Content' section with a paragraph of text. Then, a 'Student Queries' section lists three queries from students: Atharva Sadanand Litake, Apurva Ajit Kulkarni, and another Apurva Ajit Kulkarni, each with a dropdown arrow.

ADMIN
Administrator Portal

Dashboard

Add Job Opportunity

Send Notices

Listed Jobs

View Registered Students

View All Students

Mark as Placed

Mark as Shortlisted

Help & Support

Logout

Help & Support

This section is designed for resolving student queries directly by TNP, ensuring quick communication and efficient resolution of placement-related concerns.

SUMMARIZE QUERIES

Summarized Content:

Students have submitted various queries regarding registration deadlines, technical issues, and company-specific information. One student inquired about the last date to register for Energy Exemplar. Another student reported a non-functional ZS test platform, which has already been replied to. There are also questions concerning the release time of HSBC results and specific skill requirements for MasterCard. These queries indicate a need for readily available information on registration deadlines, platform functionality, and company-specific requirements to address student concerns effectively.

Student Queries:

Atharva Sadanand Litake

When is the last date to register for Energy Exemplar

▼

Apurva Ajit Kulkarni

hi my zs test platform is not working

▼

Apurva Ajit Kulkarni

When will be HSBC Results out ? Any particular time slot for it?

▼

Figure 8.24: Help and Support

CHAPTER 9

OTHER SPECIFICATIONS

9.1 Advantages

The following are the advantages of using our project.

- **Centralized Platform:** All placement related information will be available on this platform .
- **Updates and Notifications:** Instant and regular updates about company hiring process.
- **Automated Tasks:** Information verification, eligibility check for each company and resume shortlisting, etc. will be done automatically through this system.
- **Easy Communication:** Any important information and updates will be directly displayed on dashboard.
- **Mobile Access:** This application will be available on mobile and can be accessed easily.
- **Simplified Applications:** Students can easily apply for jobs which they are interested in.
- **Reduce Workload:** Will reduce work of Training and Placement Department to manage process efficiently.
- **Tracking and Reporting:** Students details and applications can be tracked easily and if there are any issues or queries , students can report them in chat section and will be clarified.
- **Consistency:** The consistent data will be maintained in system with regular updation.
- **Secure login mechanism:** Students will create there own account and login credentials will be allocated to them to maintain privacy.
- **Campus drive management:** Helps organize and manage multiple campus recruitment events.

9.2 Limitations

This project is developed by taking into consideration the requirements and issues faced during hiring process at SCTR's Pune Institute of Computer Technology. The overall structure of the project is shaped as per the requirements from higher authorities and other stakeholders. This introduces a rigidity and scalability limitation in the project. The portal cannot be versatile and suitable for hiring process of each and every organization.

9.3 Applications

- **Campus Recruitment:** Facilitates the entire placement process for on-campus hiring, job postings from various companies, shortlisting of eligible students based on their eligibility, timely notifications regarding interview schedules, and the announcement of final results. This ensures a smooth and efficient recruitment experience for both students and employers.
- **Internship Drives:** Helps students discover and apply for internship opportunities across multiple industries. The portal provides a comprehensive list of available internships, enabling students to gain valuable work experience.
- **Training Programs:** Manages student participation in various skill development and placement preparation sessions. To succeed students in their job searches, ensuring they are well-prepared for the competitive job market.
- **Industry Collaborations:** Facilitates partnerships with industries to organize specialized recruitment drives, targeted training programs, and internships. These collaborations are designed to ensure that all opportunities provided are genuine and fair, promoting transparency and integrity in the recruitment process, thereby preventing any unfair means.

CHAPTER 10

CONCLUSION & FUTURE WORK

10.1 Conclusion

In conclusion, development of a centralized automated platform for placement process proves to be effective and resolves various problems faced by both the parties. Students can now register once and apply to many job opportunities with a single click without repetitive form filling. The system provides a detailed display of all available job opportunities which effectively allows every student to explore and apply for each available opportunity. Additionally, the comprehensive chat module resolves all the issues faced by candidates throughout the process. On administrator side, this project proves to be an efficient tool for managing all student and company related affairs. Overall, the objective of this project to streamline the hiring process and reducing the workload of Training and Placement Department has been fulfilled.

10.2 Future Work

Although this project has proved to be an efficient tool for streamlining the hiring process, there are several areas in which the project could be enhanced upon in future.

- **Versatility:** The project has been developed based upon the requirements and expectations of higher authorities and stakeholders of SCTTR's Pune Institute of Computer Technology. A flexibility can be introduced in the project by taking into consideration industry standards of other organizations as well.
- **Virtual Interview Module:** Most of the companies prefer to have virtual interview rounds through various video calling services such as Google Meet, Microsoft Teams, etc. A virtual interview module can be implemented within the portal to ease out the process of virtual interview.
- **Training Modules:** Based upon candidates past online assessment results and interview reviews, appropriate training modules can be delivered to the candidate to cope up with the recent industry needs and excel in career.

CHAPTER 11

APPENDIX

11.1 Appendix A

11.1.1 Technical Feasibility

- The project leverages modern technologies like Vue ensuring technical feasibility and scalability.
- The integration of Node.js, Prisma, and PostgreSQL for backend development provides a robust and scalable environment, ensuring smooth handling of large volumes of student and employer data, demonstrating the technical viability of the platform.

11.1.2 Operational Feasibility

- User testing and feedback from hiring companies and candidates indicate that the application is operationally feasible.
- The user-friendly interface and comprehensive features of the platform are well-received, streamlining the recruitment process efficiently.

11.1.3 Economic Feasibility

- A cost-benefit analysis was conducted to assess the economic feasibility of the project.
- The costs associated with platform development, maintenance, and potential return on investment were taken into account.
- The results showed that the benefits are greater than the costs, making the project financially viable.

11.2 Appendix B

11.2.1 Survey Paper Details

Paper Status: Published

Name of Journal: Communications on Applied Nonlinear Analysis

Page Numbers: 16

DOI: <https://doi.org/10.52783/cana.v32.4113>

11.2.2 Implementation Paper

Paper Status: Submitted

Name of Journal:

Page Numbers: 6

DOI:

11.3 Appendix C

11.3.1 Plagiarism Report

The plagiarism report for the project report has been attached herewith.

Similarity: 8%

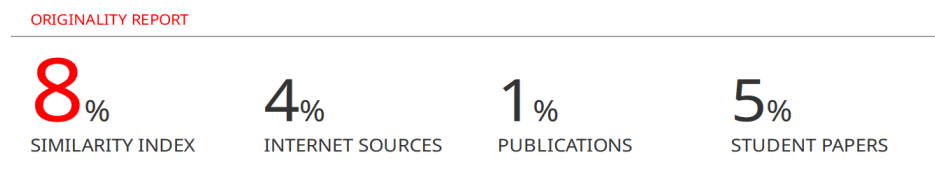


Figure 11.1: Plagiarism Report

CHAPTER 12

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
LLM	Large Language Model
ORM	Object Relational Mapping
MLOps	Machine Learning Operations
TNP	Training and Placement
AI	Artificial Intelligence
API	Application Programming Interface
E2E	End-to-End
UI	User Interface
JWT	JSON Web Token
CTC	Cost to Company